

Review/Thoracic imaging

Artificial intelligence: A critical review of applications for lung nodule and lung cancer

Constance de Margerie-Mellon^{a,b,*}, Guillaume Chassagnon^{c,d}^a Université Paris Cité, Laboratory of Imaging Biomarkers, Center for Research on Inflammation, UMR 1149, INSERM, 75018 Paris, France^b Department of Radiology, Hôpital Saint-Louis APHP, 75010 Paris, France^c Université Paris Cité, Faculté de Médecine, 75006 Paris, France^d Department of Radiology, Hôpital Cochin APHP, 75014 Paris, France

ARTICLE INFO

Keywords:

Artificial intelligence
Deep learning
Multidetector computed tomography
Pulmonary nodule
Lung neoplasms

ABSTRACT

Artificial intelligence (AI) is a broad concept that usually refers to computer programs that can learn from data and perform certain specific tasks. In the recent years, the growth of deep learning, a successful technique for computer vision tasks that does not require explicit programming, coupled with the availability of large imaging databases fostered the development of multiple applications in the medical imaging field, especially for lung nodules and lung cancer, mostly through convolutional neural networks (CNN). Some of the first applications of AI in this field were dedicated to automated detection of lung nodules on X-ray and computed tomography (CT) examinations, with performances now reaching or exceeding those of radiologists. For lung nodule segmentation, CNN-based algorithms applied to CT images show excellent spatial overlap index with manual segmentation, even for irregular and ground glass nodules. A third application of AI is the classification of lung nodules between malignant and benign, which could limit the number of follow-up CT examinations for less suspicious lesions. Several algorithms have demonstrated excellent capabilities for the prediction of the malignancy risk when a nodule is discovered. These different applications of AI for lung nodules are particularly appealing in the context of lung cancer screening. In the field of lung cancer, AI tools applied to lung imaging have been investigated for distinct aims. First, they could play a role for the non-invasive characterization of tumors, especially for histological subtype and somatic mutation predictions, with a potential therapeutic impact. Additionally, they could help predict the patient prognosis, in combination to clinical data. Despite these encouraging perspectives, clinical implementation of AI tools is only beginning because of the lack of generalizability of published studies, of an inner obscure working and because of limited data about the impact of such tools on the radiologists' decision and on the patient outcome. Radiologists must be active participants in the process of evaluating AI tools, as such tools could support their daily work and offer them more time for high added value tasks.

© 2022 Société française de radiologie. Published by Elsevier Masson SAS. All rights reserved.

1. Introduction

Artificial intelligence (AI) is a term first coined in the mid 1950's [1]. AI was then defined it as “the science and engineering of making intelligent machines, especially intelligent computer programs”. Today, AI is a broad concept without consensual definition. It usually refers to computer programs that can learn from data and perform certain specific tasks. While the concept is not new, AI has known dramatical developments in the recent years, including in the field of medical imaging [2].

Abbreviations: AI, Artificial intelligence; AUC, Area under the curve; CADe, Computed aided detection; CADx, Computer aided diagnosis; CNN, Convolutional neural network; CT, Computed tomography; DL, Deep learning; EGFR, Epidermal growth factor receptor; NLST, National Lung Cancer Screening Trial

* Corresponding author: constance.de-margerie@aphp.fr

This AI boom in radiology relies on two main pillars. The first pillar is the growth of deep learning (DL), a subfield of AI. Contrary to traditional machine learning approaches, which require hand-engineered feature extraction from images and human-based data structuration, DL-based algorithms are fed with raw data and develop their own representations needed for pattern recognition [3]. DL systems can transform input images into useful outputs, such as object locations (for detection), pixel labels (for segmentation) and image categories (for classification) [4]. Within DL, convolutional neural network (CNN) is the dominant architecture for medical image analysis. A CNN is built with multiple layers of convolutions that are arranged sequentially, such that the representation of one layer (the first being the raw data input) is fed into the next layer and transformed into a more abstract representation [3,5]. Boosted by the increase in computational power, with the advent of powerful graphics processing unit, DL is now the method of choice for medical

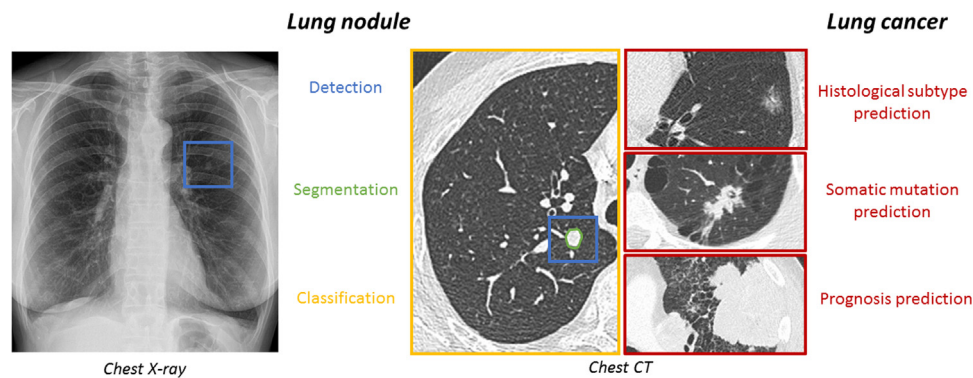


Fig. 1. Graphical representation of deep learning applications for lung nodule and lung cancer in chest imaging.

images analysis. The second pillar of the boom of AI in radiology is the availability of large image datasets thanks to the development of picture archiving and communication systems. Large image datasets are essential to train DL models.

Among all medical imaging subfields, thoracic imaging is probably the one where AI applications have been the most studied [6]. Although many respiratory diseases have been investigated [7–9], the majority of current applications of AI are related to lung nodule and lung cancer (Fig. 1).

The purpose of this article was to review and discuss the current and future applications of AI in the elective field of lung nodule and lung cancer.

2. Lung nodule

The identification of lung nodules on chest computed tomography (CT) is increasingly common. In a US-based study, Gould et al. identified both an increase of the lung nodule detection rate per 1000 person-years from 3.9 to 6.6 and of the lung nodule identification rate per 100 chest CTs from 24 to 31 between 2006 and 2012 [10]. Most of these nodules are of benign nature: in the National Lung Cancer Screening Trial (NLST), 24.2% of participants had a least one nodule over 3 mm but only 3.6% of the detected nodules were malignant [11]. However, lung cancer is still the deadliest cancer worldwide and early detection offers a unique opportunity for curative treatment [12], while 8.9% of the lung cancers were missed on initial CT in the NLST [13]. Therefore, there is a need for both avoiding missed lung nodules and classifying them between benign and malignant. AI tools offer encouraging perspectives for these goals.

2.1. Lung nodule detection

2.1.1. Chest X-ray

Chest X-ray is significantly less sensitive than chest CT for the identification of lung nodules and is not recommended for lung cancer screening [11,14]. However, it is largely used as a first-line imaging examination for respiratory diseases because of its widespread availability and its low cost [15]. Therefore, chest X-ray can offer an opportunity for nodule diagnosis that should not be overlooked. If small lung nodules are clearly undetectable on chest X-rays, it is not uncommon to identify previous significant radiological abnormalities in patients in whom a diagnosis of lung cancer is later made [16]. Observer errors are the main cause of missed lung cancers on chest X-ray examination [17]. Such errors are more usual among less experienced radiologists but can also occur among specialists in chest imaging [18,19].

For this reason, automated detection of lung nodules on chest X-ray has been a topic of interest for researchers in the last 30 years. The first conventional computer aided detection (CADe) systems employed traditional image analysis techniques with the need to

perform several successive steps (image preprocessing, lung field segmentation, feature analysis) [20]. Such CADe systems were limited by suboptimal diagnostic performances and a high rate of false positives findings; they have been scarcely implemented for clinical practice [21–23]. In the recent years, many DL-based algorithms have been developed (Fig. 2). These algorithms mostly rely on pretrained deep neural networks with the use of transfer learning, where knowledge derived from distinct but similar problems is used [20]. Large public datasets are available, the larger one being the Chest X-ray14, composed of 112 120 frontal chest X-rays labelled with 14 common pathologies including lung mass and lung nodule [24].

Nodule detection on chest X-ray with DL techniques has been developed in narrow and comprehensive models [25]. Narrow models are trained for the detection of a single finding (namely lung nodule detection), whereas comprehensive models include the detection of several abnormalities. In a 2021 meta-analysis, Aggarwal et al. obtained a diagnostic accuracy of 0.884 for the detection of lung nodules on chest X-rays, based on nine studies with narrow models, in comparison to healthcare professionals [6]. In one of the studies included in the meta-analysis, Nam et al. built a DL-based algorithm that significantly outperformed most of the physicians, for both a classification task (X-ray with a nodule versus X-ray without a nodule) and a detection task (identification of the nodule on the image) [26]. Among more recent studies, some assessed the added value of commercially available DL tools for the detection of lung cancer when used as second readers. Jang et al. showed that the average area under the curve (AUC) for all observers significantly rose from 0.67 without the DL aided-detection, to 0.76 with it, without increasing the number of chest CT recommendations in the healthy group [27]. In another dataset, Yoo et al. showed 10% increase in sensitivity for cancer detection among residents, whereas the sensitivity was similar among senior radiologists, with and without DL [28]. To tackle the issue of test datasets composed of a high proportion of chest X-rays with the disease (nodule or cancer) that do not reflect clinical practice, Lee et al. tested a commercial DL algorithm in more than 50,000 examinations, with a 0.2% prevalence of lung cancer. The model showed comparable sensitivity and negative predictive value to that of radiologists, with only 3% of chest radiographs having potentially positive results [29].

In focusing on the presence or absence of the finding they were trained to detect, narrow models may distract the radiologist from other significant abnormalities. Comprehensive models, also called multiclassifiers, go past this limitation. Lung nodule or mass detection are usually included in the detected abnormalities. Rajpundkar et al. studied 14 different pathologies, and showed similar performances of their DL algorithm, as compared to radiologists, for the detection of lung nodules and masses [30]. Using a commercially available DL algorithm, Homayounieh et al. assessed five findings and demonstrated a 10% increase in sensitivity for nodule detection with AI-aided interpretation, which was significantly higher for junior

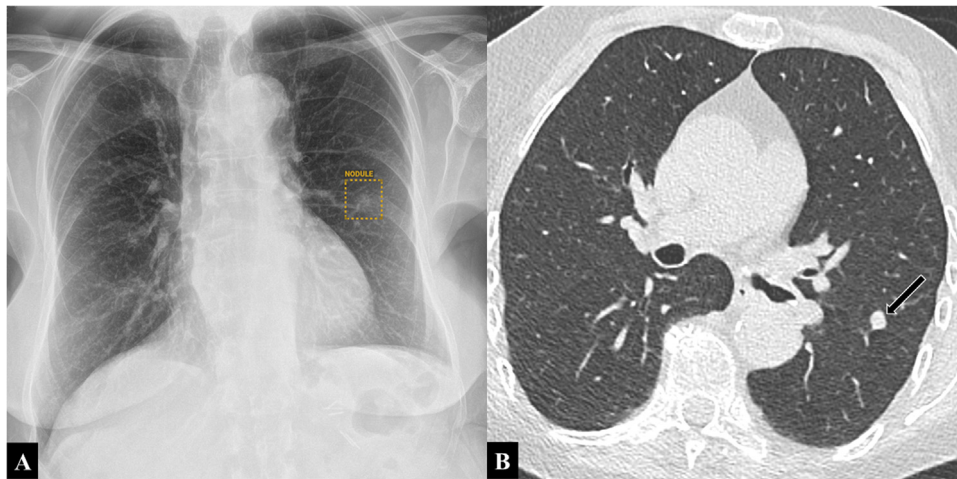


Fig. 2. A, A commercially available deep learning based software was applied on frontal chest X-ray image and identified a faintly visible nodule in the left hemithorax (square). B, Corresponding chest CT image in the axial plane confirms lung nodule located in the left lower lobe (arrow).

radiologists as compared with their senior counterparts [31]. Finally, in a comprehensive study that included 127 clinical findings, Seah et al. trained a DL algorithm on a dataset of more than 800,000 images, specifically labelled for the study needs. In an external validation cohort, the authors showed a significant increase of classification accuracy when radiologists used the DL algorithm for 102 of the 127 clinical findings, including for lung nodules [32]. Such comprehensive models are closer to clinical practice, where radiologists need to detect all visible abnormalities on the chest X-ray and may represent a significant added value.

2.1.2. Chest CT

Chest CT is a highly performant method for lung nodule detection. However, detecting lung nodules on chest CTs is a repetitive and fastidious task, that can be challenging when the nodule is small, faint or of endobronchial and hilar locations [17]. Moreover, most radiologists experience an increase in their daily workload [33] and development of lung cancer screening may further boost the number of chest CTs that are performed, while higher diagnostic examination volumes are associated with increased errors by radiologists [34]. In that context, automated detection methods for lung nodules are highly desirable.

Automatic detection of lung nodules with CT has been studied since the 1990's. As for chest X-rays, the first developed CADe tools relied on hand-engineered approaches, where each stage was explicitly programed [35]. If some of these traditional CADe tools have shown their abilities to improve the radiologist's performance for detection lung nodules when used as a second reader, they were usually associated with a high number of false positive nodules and did not find their place in the daily workflow [36]. DL-based approaches for lung nodule detection on CT have now emerged (Fig. 3). Eliminating the need for complex human-led pipelines, they are gradually becoming the most used method, with an overall higher accuracy and fewer false positives [37]. However, application of CNNs to CT images is more challenging than in chest X-rays, given the 3D nature of CT examination and the high number of images [38]. Most algorithms use two-step strategies, with a first step consisting in detecting candidate nodules with a high specificity, followed by a false positive reduction step where candidate nodules are classified as true or false nodules. Several open databases containing annotated CT examinations are available for researchers to train and test their algorithms dedicated to lung nodule detection and classification, such as the LIDC-IDRI and the NLST datasets [11,39].

In 2016, in one of the first studies using CNN, Setio et al. obtained 85.4% sensitivity with one false positive per examination [40].

Sensitivity could be improved to 90.1% at the cost of four false positives per examination. Including only nodules ≥ 5 mm, Huang et al. lowered the average number of false positives per examination to 0.125 and 0.25, for 81.7% and 85.1% sensitivity, respectively [41]. A recent meta-analysis assessed the performances of lung nodule detection on CT [6]. Among 56 studies with 74 separate patient cohorts, the reported AUC for identifying lung nodules on CT examination was 0.937 on a per lesion basis [6]. However, such results were not compared to a radiologist's performance. For that aim, Li et al. tested a commercial DL-based CADe tool in 346 lung cancer screening participants [42]. The DL CADe model had a significantly higher detection rate of lung nodules than that of double reading by two thoracic radiologists (86.2% vs. 79.2%) with a false positive rate of 1.53 per CT examination. However, this false positive rate was still higher compared to that of the radiologists (0.13 per CT examination). Other commercially DL-based programs are now available for lung nodule detection, to be used as a second or concurrent reader [43,44], but further research to reduce the false positive rate is still warranted.

Of note, subsolid nodules are sometimes excluded from nodule detection studies. Subsolid nodules, that include both pure ground-glass nodules and part-solid nodules, are clinically important because they have an higher rate of malignancy, as compared to solid nodules [45], but as low-contrast lesions, there are more challenging to detect. Using traditional CADe systems, Jacobs et al. and Silva et al. showed that these algorithms could reach sensitivities of 80 to 80.4% [46,47] but using DL-based tools, higher performances could be reached. Park et al. reached 92.2% sensitivity, whereas Singh et al. showed that subsolid nodule detection was improved when using an AI-based tool for vessel suppression [48,49].

2.2. Lung nodule segmentation

Segmentation can be considered as a voxel-level classification task, where each voxel of the image is categorized as belonging to the nodule or the background [50]. Reliable segmentation allows accurate volume estimation, which is necessary for nodule management, especially for volume doubling time computation [51]. Manual segmentation is time consuming and can be prone to observer variability. Therefore, accurate automated segmentation is of high interest for clinical practice.

Multiple commercial software that provide automated or semi-automated nodule segmentation are now available, most of them also offering automated detection of lung nodules prior to volume estimation. Traditional segmentation algorithms were mainly based

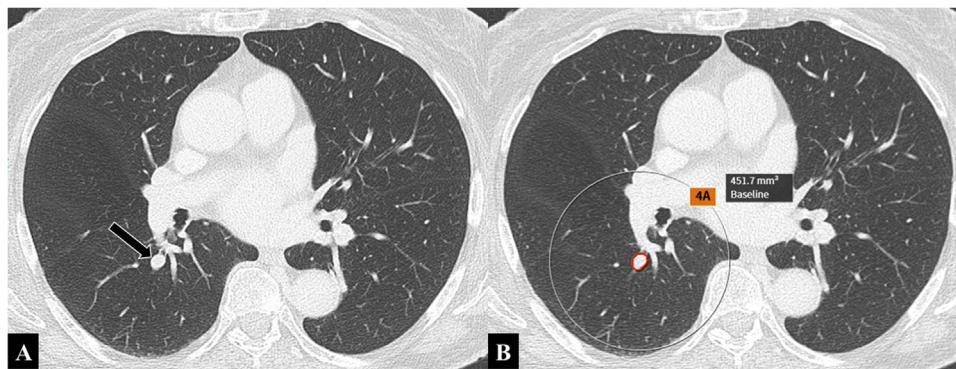


Fig. 3. A, Chest CT image in the axial plane shows a challenging nodule in the right perihilar area (arrow). B, A commercially available deep-learning based software was applied on the CT volume and was able to identify the nodule. The nodule was correctly classified as Lung-RADS 4a, based on its dimension.

on attenuation thresholding, to isolate the nodule from its aerated environment, followed by morphological processing, to eliminate possible adjacent vessels or subpleural regions [52]. Such software usually show weaknesses for nodules with irregular shapes and sub-solid nodules [53]. Because of these limitations, CNN-based segmentation algorithms have become the standard approach for medical image segmentation. Such algorithms usually rely on an encoder-decoder architecture.

Studies dedicated to segmentation usually use the DICE similarity coefficient as a metrics to assess the performances of the model. The DICE similarity coefficient measures the spatial overlap between the segmentation and the ground truth. High DICE similarity coefficients, over 92%, were recently obtained for the segmentation of lung nodules [54,55].

2.3. Lung nodule classification

Once a nodule is detected and measured, an appropriate management can be decided based on current guidelines [51,56,57]. In general, small nodules will be neglected, while patients with highly suspicious nodules (because of their size and/or their morphological characteristics) will undergo histological sampling, PET-CT and tumor board referral. In between, indeterminate nodules, defined by an intermediate size (usually 6–8 mm, 80 to 300 mm³), will be followed up with CT. Indeterminate nodules are not uncommon, and most of them are of benign nature: for instance, in the NLST, the rate of false positive screening tests (patients with at least one nodule ≥ 4 mm) was 96.4% [11]. CT follow-up creates significant cost, additional radiation exposure, and potential anxiety for the patient. Therefore, a more accurate estimation of lung cancer risk for a given nodule would be valuable, especially at the time when lung cancer screening becomes widespread.

Several models that rely on both clinical (sex, gender, smoking habits) and simple CT features (nodule size, density, location, borders) have been developed such as the Brock model, the VA model or the Swenson model, based on robust historical cohorts [58]. These models provide a more accurate and quantitative estimation of lung cancer risk than nodule size only and can be helpful to personalize nodule management. Such models could perform similarly to clinical judgement of experienced radiologists, with AUC for predicting malignancy up around 0.8 [59]. Going further in lung nodule characterization, to, ultimately, alleviate the need of CT follow-up in nodules identified as benign, more sophisticated quantitative models have been explored and are often named CADx (for computer aided diagnosis) tools. The first quantitative studies were based on radiomics. Radiomics use an explicit multistep method: nodules are first segmented, then predefined quantitative features (related to the morphology, the density and the texture) are extracted, and finally an optimal model (ie the radiomics signature) is built, based on the most relevant features among included nodules, using classical

machine learning algorithms [60]. Results are encouraging, with reported AUCs over 0.9 for the prediction of malignant nodules in some studies [61–63]. While radiomics has the advantage of explainability because the user knows which features have brought the algorithm to its conclusion, it also has drawbacks. Among them, the feature extraction step, which is cumbersome and may be influenced by technical factors related to acquisition and reconstruction parameters, inducing potential variability that is independent of the studied question [64].

Many classification tools are now DL-based. CNNs with different architectures have been used successfully; many tools offer first a detection step followed by a classification step limited on a pre specified region of interest [65]. In 2019, Google AI and collaborators developed a DL tool based on a 3D CNN architecture that used a patient's current and prior entire chest CT volumes to predict the risk of lung cancer within 1 year. Their model trained with NLST data achieved an AUC of 0.955 on an independent validation set. The algorithm performance was also compared to that of 6 radiologists. When a prior CT was not available, the model outperformed radiologists with absolute reductions of 11% in false positive findings and 5% in false negative findings; if prior CT was available, the model performance was similar to that of the radiologists [66]. A limitation of this study was that nodule assessment by radiologists was based on Lung RADS score, which provide management categories rather than a true risk cancer estimation [57]. Using a different approach, Huang et al. applied a DL algorithm not on the image themselves, but on clinical features and on nodule and non-nodule features identified by a radiologist on two successive lung cancer screening CT examination [67]. Their algorithm could predict the presence of lung cancer within a three-year period, with a 1-, 2- and 3-year time-dependent AUC values of 0.968, 0.946 and 0.899. Such results could open the way to tailored CT intervals in lung cancer screening programs. Finally, recently, Jacobs et al. assessed the 3 top performing algorithms from the 2017 Kaggle Data Science Bowl, a public competition which aim was the assessment of the risk of lung cancer at 1 year. A dataset composed of 300 CT examinations (half of them containing malignant nodules) was used for the challenge. Model outputs were compared with an observer study of 11 radiologists who assessed the same test datasets. The scoring was based on a continuous scale for both the DL algorithms and the radiologists. The radiologists' average AUC was 0.917 and was not significantly different from the one obtained for two out of three DL-based algorithms [68]. A limitation of these algorithms was that they only provided an overall risk cancer score for the CT examination without localization of the cancer.

3. Lung cancer

Lung cancer is the leading cause of cancer death worldwide [69]. However, lung cancer mortality is gradually decreasing, because of a

slow decline in incidence but also likely because of treatment improvement, especially with the advent of targeted therapies and immune checkpoint inhibitors [70]. In lung cancer, AI could be used to select the optimal management by allowing noninvasive staging and mutation and prognosis prediction with minimal additional cost, as CT examinations are performed as part of the standard-of-care practice.

When a lung cancer is suspected, staging is capital to select the appropriate treatment. In the case of pulmonary adenocarcinomas presenting as subsolid nodules, identifying the degree of invasiveness can be important to guide surgery. Numerous studies focused on this challenge, first using simple morphological patterns, and then with radiomics analysis, with encouraging results but the above-mentioned limitations of radiomics [71–73] namely the segmentation step burden, and the pitfall of technical variability. In two recent studies, Huang et al. and Gong et al. used DL for this task and obtained AUCs of 0.957 and 0.96, respectively [74,75]. Interestingly, Huang et al. incorporated not only radiomics features but also clinical and biological markers in their DL model. Other researchers reported the use of DL tools to predict N2 status in non-small cell lung cancer [76].

Mutation prediction is another interesting field of research in lung cancer imaging. With the development of targeted therapies, non-invasive identification of patients with tumor somatic mutations can be critical, to avoid lung biopsies and the risk of inadequate tissue sampling. In a cohort including 353 lung adenocarcinomas, a radiomics-based method achieved an AUC of 0.69 by using hand-crafted CT image features to predict epidermal growth factor receptor (EGFR) phenotype [77]. In 2019, Wang et al. built an effective DL model for the prediction of EGFR mutation from CT images, reaching an higher AUC of 0.81 while limiting the need for human input compared to the radiomics method [78]. In 2022, the same team went further by mining whole-lung information from CT images to predict EGFR phenotype, instead of focusing only on the tumor characteristics. This model achieved AUCs ranging from 0.748 to 0.813 in the different testing cohorts [79]. The prediction of tumor mutational burden, a significant predictor of immune checkpoint inhibitors efficacy, has also been studied with DL applied to CT images. He et al. obtained an AUC of 0.81 to discriminate between lesions with low and high-tumor mutational burdens [80].

Finally, some researchers worked on the potential impact of DL-algorithms to predict prognosis. In a multicenter study based on 1194 NSCLC patients, Hosny et al. found that a CNN model was able to stratify patients into low and high mortality risk groups after radiotherapy and surgery. In patients treated with surgery, the DL algorithm significantly outperformed models based on predefined tumor features as well as tumor volume and maximum diameter [81]. In a dataset of 179 patients with stage III non-small cell lung cancers treated with definitive chemoradiation, with pretreatment and post treatment CT images at 1-, 3-, and 6-months follow-up, Kim et al. showed that DL models based on time series CT examinations were significantly predictive of survival and cancer-specific outcomes (progression, distant metastases, and local-regional recurrence). Model performance was enhanced with each additional follow-up CT examination included into the model, with an AUC of 0.74 for the 2-year overall survival [82].

4. Current limitations and future directions

AI, mostly through DL, offers new horizons in lung nodule and lung cancer imaging. Interesting tools that could support and strengthen the radiologist's work are being developed. Such tools are especially desirable at the time where lung cancer screening programs with CT are being implemented worldwide, with a potential increased workload for radiologists, and a specific need for a better identification of suspicious nodules that could limit the rate of false positives and allow personalized screening intervals. They could also

be of high interest for junior radiologists who have a limited clinical experience. However, despite a high number of publications dedicated to DL in the lung nodule and lung cancer fields, clinical implementation is slow for several reasons [83].

First, published studies are deeply heterogeneous, and synthesizing their results, to increase the strength of each author's claim, is challenging [6]. Datasets, algorithms architectures, reference standards and performance metrics vary a lot across publications, making comparisons difficult. Many studies do not include external validation in independent datasets, and/or do not perform a comparison to radiologists' performances, whereas it is a major issue [84]. Thus, generalizability and applicability of reported DL algorithms can be questioned. The Radiology AI check list, recommended for authors, can also be helpful for readers when critically reviewing DL-based studies [85]. Second, regarding predictive models (e.g., nodule classification, somatic mutation or prognosis prediction), the clinical implementation is limited by the obscure inner working (often called "black box") and the lack of theoretical backing [86]. To be confident about an AI tool, the user needs a clear understanding of its functioning. Third, it is difficult to determine what threshold of performance is acceptable for clinical use, both for detection and classification tasks. Even if the algorithm reaches a diagnostic accuracy that is non-inferior to radiologists, additional value must be demonstrated, in terms of speed, efficiency and/or cost [6]. Finally, the impact of AI tools on radiologists' decision is mostly unknown, as well as the consequences for the patient care and outcome [87].

To date, there are several commercial AI programs for lung nodule detection (on chest X-ray and on chest CT) and segmentation (on chest CT) [88]. A comprehensive overview of currently available AI software can be consulted on the website *AIforRadiology.com*, which collects software characteristics and associated publications, with the aim of increasing transparency in this domain. On the other hand, commercial software for classification is still lacking, for the reasons developed above. While there is no doubt that DL has great abilities in the field of medical imaging, further work is still needed at distinct levels, regarding the application of standardization guidelines for building and reporting DL-based studies, the availability of larger, high-quality datasets, the evaluation in terms of clinical outcome of AI tools that are already commercially available, the adaptation of legal and ethical frameworks with the issue of liability [89]. Most of all, radiologists, who may have the feeling of receiving insufficient information about AI must take a central role in DL tools development and implementation, by identifying the optimal workflows for their use [90]. With the support of DL, radiologists could thus use the time gain for higher added values tasks, such as integration of the radiological findings in the overall clinical management, going beyond the mere image analysis [91].

Human rights

The authors declare that the work described has been performed in accordance with the Declaration of Helsinki of the World Medical Association revised in 2013 for experiments involving humans.

Informed consent and patient details

The authors declare that this article does not contain any personal information that could lead to the identification of the patients.

Disclosure of interest

The authors declare that they have no competing financial or personal relationships that could be viewed as influencing the work reported in this paper.

Funding

This work did not receive any grant from funding agencies in the public, commercial, or not-for-profit sectors.

Author contributions

All authors attest that they meet the current International Committee of Medical Journal Editors (ICMJE) criteria for Authorship.

References

- [1] McCarthy J, Minsky ML, Rochester N, Shannon CE. A proposal for the Dartmouth summer research project on artificial intelligence. *AI Mag* 2006;27:12–4.
- [2] Tang X. The role of artificial intelligence in medical imaging research. *BJR Open* 2020;2:20190031.
- [3] Esteve A, Robicquet A, Ramsundar B, Kuleshov V, DePristo M, Chou K, et al. A guide to deep learning in healthcare. *Nat Med* 2019;25:24–9.
- [4] Nakaura T, Higaki T, Awai K, Ikeda O, Yamashita Y. A primer for understanding radiology articles about machine learning and deep learning. *Diagn Interv Imaging* 2020;101:765–70.
- [5] Blanc D, Racine V, Khalil A, Deloche M, Broyelle JA, Hammouamri I, et al. Artificial intelligence solution to classify pulmonary nodules on CT. *Diagn Interv Imaging* 2020;101:803–10.
- [6] Aggarwal R, Sounderajah V, Martin G, Ting DSW, Karthikesalingam A, King D, et al. Diagnostic accuracy of deep learning in medical imaging: a systematic review and meta-analysis. *NPJ Digit* 2021;4:65.
- [7] Iqbal T, Shaikat A, Akram MU, Mustansar Z, Khan A. Automatic diagnosis of pneumothorax from chest radiographs: a systematic literature review. *IEEE Access* 2021;9:145817–39.
- [8] Kato S, Ishiwata Y, Aoki R, Iwasawa T, Hagiwara E, Ogura T, Utsunomiya D. Imaging of COVID-19: an update of current evidences. *Diagn Interv Imaging* 2021;102:493–500.
- [9] Li J, Long X, Wang X, Fang F, Lv X, Zhang D, et al. Radiology indispensable for tracking COVID-19. *Diagn Interv Imaging* 2021;102:69–75.
- [10] Gould MK, Tang T, Liu IL, Lee J, Zheng C, Danforth KN, et al. Recent trends in the identification of incidental pulmonary nodules. *Am J Respir Crit Care Med* 2015;192:1208–14.
- [11] Aberle DR, Adams AM, Berg CD, Black WC, Clapp JD, Fagerstrom RM, et al. Reduced lung-cancer mortality with low-dose computed tomographic screening. *N Engl J Med* 2011;365:395–409.
- [12] Barta JA, Powell CA, Wisnivesky JP. Global epidemiology of lung cancer. *Ann Glob Health* 2019;85:8.
- [13] Scholten ET, Horeweg N, de Koning HJ, Vliegenthart R, Oudkerk M, Mali WP, et al. Computed tomographic characteristics of interval and post screen carcinomas in lung cancer screening. *Eur Radiol* 2015;25:81–8.
- [14] Oken MM, Hocking WG, Kvale PA, Andriole GL, Buys SS, Church TR, et al. Screening by chest radiograph and lung cancer mortality: the prostate, lung, colorectal, and ovarian (PLCO) randomized trial. *JAMA* 2011;306:1865–73.
- [15] (UNSCLEAR) UNSCotEoAR. Sources, effects and risks of ionizing radiation - Annex A: evaluation of medical exposure to ionizing radiation, https://www.unsclar.org/unsclar/publications/2020_2021_1.html, 2021 (accessed 10.10.22)
- [16] Turkington PM, Kennan N, Greenstone MA. Misinterpretation of the chest x ray as a factor in the delayed diagnosis of lung cancer. *Postgrad Med J* 2002;78:158–60.
- [17] del Ciello A, Franchi P, Contegiacomo A, Cicchetti G, Bonomo L, Larici AR. Missed lung cancer: when, where, and why? *Diagn Interv Radiol* 2017;23:118–26.
- [18] Brogdon BG, Kelsey CA, Moseley RD. Factors affecting perception of pulmonary lesions. *Radiol Clin North Am* 1983;21:633–54.
- [19] Kundel HL, La Follette Jr PS. Visual search patterns and experience with radiological images. *Radiology* 1972;103:523–8.
- [20] Moses DA. Deep learning applied to automatic disease detection using chest X-rays. *J Med Imaging Radiat Oncol* 2021;65:498–517.
- [21] Shiraishi J, Li Q, Suzuki K, Engelmann R, Doi K. Computer-aided diagnostic scheme for the detection of lung nodules on chest radiographs: localized search method based on anatomical classification. *Med Phys* 2006;33:2642–53.
- [22] Hardie RC, Rogers SK, Wilson T, Rogers A. Performance analysis of a new computer aided detection system for identifying lung nodules on chest radiographs. *Med Image Anal* 2008;12:240–58.
- [23] Schilham AMR, van Ginneken B, Loog M. A computer-aided diagnosis system for detection of lung nodules in chest radiographs with an evaluation on a public database. *Med Image Anal* 2006;10:247–58.
- [24] Wang XS, Peng YF, Lu L, Lu ZY, Bagheri M, Summers RM, et al. ChestX-ray8: hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. In: 30th IEEE/CVF Conference on Computer Vision and Pattern Recognition; 2017. p. Honolulu, HI3462–671.
- [25] Jones CM, Buchlak QD, Oakden-Rayner L, Milne M, Seah J, Esmaili N, et al. Chest radiographs and machine learning: past, present and future. *J Med Imaging Radiat Oncol* 2021;65:538–44.
- [26] Nam JG, Park S, Hwang EJ, Lee JH, Jin KN, Lim KY, et al. Development and validation of deep learning-based automatic detection algorithm for malignant pulmonary nodules on chest radiographs. *Radiology* 2019;290:218–28.
- [27] Jang S, Song H, Shin YJ, Kim J, Lee KW, Lee SS, et al. Deep learning-based automatic detection algorithm for reducing overlooked lung cancers on chest radiographs. *Radiology* 2020;296:652–61.
- [28] Yoo H, Lee SH, Arru CD, Khera RD, Singh R, Siebert S, et al. AI-based improvement in lung cancer detection on chest radiographs: results of a multi-reader study in NLST dataset. *Eur Radiol* 2021;31:9664–74.
- [29] Lee JH, Sun HY, Park S, Kim H, Hwang EJ, Goo JM, et al. Performance of a deep learning algorithm compared with radiologic interpretation for lung cancer detection on chest radiographs in a health screening population. *Radiology* 2020;297:687–96.
- [30] Rajpurkar P, Irvin J, Ball RL, Zhu KL, Yang B, Mehta H, et al. Deep learning for chest radiograph diagnosis: a retrospective comparison of the CheXNeXt algorithm to practicing radiologists. *PLoS Med* 2018;15:e1002686.
- [31] Homayounieh F, Digumarthy S, Ebrahimian S, Rueckel J, Hoppe BF, Sabel BO, et al. An artificial intelligence-based chest X-ray model on human nodule detection accuracy from a multicenter study. *Jama Netw Open* 2021;4:e2141096.
- [32] Seah JCY, Tang CHM, Buchlak QD, Holt XG, Wardman JB, Aimoldin A, et al. Effect of a comprehensive deep-learning model on the accuracy of chest x-ray interpretation by radiologists: a retrospective, multireader multicase study. *Lancet Digit Health* 2021;3:e496–506.
- [33] McDonald RJ, Schwartz KM, Eckel LJ, Diehn FE, Hunt CH, Bartholmai BJ, et al. The effects of changes in utilization and technological advancements of cross-sectional imaging on radiologist workload. *Acad Radiol* 2015;22:1191–8.
- [34] Hanna TN, Lamoureux C, Krupinski EA, Weber S, Johnson JO. Effect of shift, schedule, and volume on interpretive accuracy: a retrospective analysis of 2.9 million radiologic examinations. *Radiology* 2018;287:205–12.
- [35] Halder A, Dey D, Sadhu AK. Lung nodule detection from feature engineering to deep learning in thoracic CT images: a comprehensive review. *J Digit Imaging* 2020;33:655–77.
- [36] Goo JM. A computer-aided diagnosis for evaluating lung nodules on chest CT: the current status and perspective. *Korean J Radiol* 2011;12:145–55.
- [37] Tandon YK, Bartholmai BJ, Koo CW. Putting artificial intelligence (AI) on the spot: machine learning evaluation of pulmonary nodules. *J Thorac Dis* 2020;12:6954–65.
- [38] Ather S, Kadir T, Gleeson F. Artificial intelligence and radiomics in pulmonary nodule management: current status and future applications. *Clin Radiol* 2020;75:13–9.
- [39] Armato SG, McLennan G, Bidaut L, McNitt-Gray MF, Meyer CR, Reeves AP, et al. The lung image database consortium and image database resource initiative: a completed reference database of lung nodules on CT. *Med Phys* 2011;38:915–31.
- [40] Setio AAA, Ciompi F, Litjens G, Gerke P, Jacobs C, van Riel SJ, et al. Pulmonary nodule detection in CT images: false positive reduction using multi-view convolutional networks. *IEEE Trans Med Imaging* 2016;35:1160–9.
- [41] Huang WK, Xue YH, Wu Y. A CAD system for pulmonary nodule prediction based on deep three-dimensional convolutional neural networks and ensemble learning. *PLoS One* 2019;14:e0219369.
- [42] Li L, Liu Z, Huang H, Lin M, Luo DH. Evaluating the performance of a deep learning-based computer-aided diagnosis (DL-CAD) system for detecting and characterizing lung nodules: comparison with the performance of double reading by radiologists. *Thorac Cancer* 2019;10:183–92.
- [43] Mueller FC, Raaschou H, Akhtar N, Brejnebol M, Collatz L, Andersen MB. Impact of concurrent use of artificial intelligence tools on radiologists reading time: a prospective feasibility study. *Acad Radiol* 2022;29:1085–90.
- [44] Martins Jarnalo CO, Linsen PVM, Blazis SP, van der Valk PHM, Dickerscheid DBM. Clinical evaluation of a deep-learning-based computer-aided detection system for the detection of pulmonary nodules in a large teaching hospital. *Clin Radiol* 2021;76:838–45.
- [45] Henschke CI, Yankelevitz DF, Mirtcheva R, McGuinness G, McCauley D, Miettinen OS. CT screening for lung cancer: frequency and significance of part-solid and nonsolid nodules. *AJR Am J Roentgenol* 2002;178:1053–7.
- [46] Silva M, Schaefer-Prokop CM, Jacobs C, Capretti G, Ciompi F, van Ginneken B, et al. Detection of subsolid nodules in lung cancer screening: complementary sensitivity of visual reading and computer-aided diagnosis. *Invest Radiol* 2018;53:441–9.
- [47] Jacobs C, van Rikxoort EM, Twellmann T, Scholten ET, de Jong PA, Kuhnigk JM, et al. Automatic detection of subsolid pulmonary nodules in thoracic computed tomography images. *Med Image Anal* 2014;18:374–84.
- [48] Park S, Lee SM, Kim W, Park H, Jung KH, Do KH, et al. Computer-aided detection of subsolid nodules at chest CT: improved performance with deep learning-based CT section thickness reduction. *Radiology* 2021;299:211–9.
- [49] Singh R, Kalra MK, Homayounieh F, Nitiwarangkul C, McDermott S, Little BP, et al. Artificial intelligence-based vessel suppression for detection of sub-solid nodules in lung cancer screening computed tomography. *Quant Imaging Med Surg* 2021;11:1134–43.
- [50] Chartrand G, Cheng PM, Vorontsov E, Drozdal M, Turcotte S, Pal CJ, et al. Deep learning: a primer for radiologists. *Radiographics* 2017;37:2113–31.
- [51] Callister ME, Baldwin DR, Akram AR, Barnard S, Cane P, Draffan J, et al. British Thoracic Society guidelines for the investigation and management of pulmonary nodules. *Thorax* 2015;70 ii1–ii54.
- [52] Gu D, Liu G, Xue Z. On the performance of lung nodule detection, segmentation and classification. *Comput Med Imaging Graph* 2021;89:101886.
- [53] Devaraj A, van Ginneken B, Nair A, Baldwin D. Use of volumetry for lung nodule management: theory and practice. *Radiology* 2017;284:630–44.
- [54] Kumar S, Raman S. Lung nodule segmentation using 3-dimensional convolutional neural networks. *Soft computing for problem solving. Socpro* 2018;585–96 Vol 1 2020;1048.

- [55] Dong X, Xu S, Liu Y, Wang A, Saripan MI, Li L, et al. Multi-view secondary input collaborative deep learning for lung nodule 3D segmentation. *Cancer Imaging* 2020;20:53.
- [56] MacMahon H, Naidich DP, Goo JM, Lee KS, Leung ANC, Mayo JR, et al. Guidelines for management of incidental pulmonary nodules detected on CT images: from the Fleischner Society 2017. *Radiology* 2017;161:659.
- [57] A.C.O. Radiology, lung CT screening reporting and data system (Lung-RADS) version 2022. <https://www.acr.org/-/media/ACR/Files/RADS/Lung-RADS/Lung-RADS-2022.pdf> (accessed 15.11.22).
- [58] Fox AH, Tanner NT. Approaches to lung nodule risk assessment: clinician intuition versus prediction models. *J Thorac Dis* 2020;12:3296–302.
- [59] Tanner NT, Porter A, Gould MK, Li XJ, Vachani A, Silvestri GA. Physician assessment of pretest probability of malignancy and adherence with guidelines for pulmonary nodule evaluation. *Chest* 2017;152:263–70.
- [60] Gillies RJ, Schabath MB. Radiomics improves cancer screening and early detection. *Cancer Epidemiol Biomarkers Prev* 2020;29:2556–67.
- [61] Peikert T, Duan FH, Rajagopalan S, Karwowski RA, Clay R, Robb RA, et al. Novel high-resolution computed tomography-based radiomic classifier for screen-identified pulmonary nodules in the national lung screening trial. *PLoS One* 2018;13:e0196910.
- [62] Mao LT, Chen H, Liang MZ, Li KW, Gao JB, Qin PX, et al. Quantitative radiomic model for predicting malignancy of small solid pulmonary nodules detected by low-dose CT screening. *Quant Imaging Med Surg* 2019;9:263–72.
- [63] Choi W, Oh JH, Riyahi S, Liu CJ, Jiang F, Chen WE, et al. Radiomics analysis of pulmonary nodules in low-dose CT for early detection of lung cancer. *Med Phys* 2018;45:1537–49.
- [64] Lee JH, Hwang EJ, Kim H, Park CM. A narrative review of deep learning applications in lung cancer research: from screening to prognostication. *Transl Lung Cancer Res* 2022;11:1217–29.
- [65] Li D, Mikela Vilmun B, Frederik Carlsen J, Albrecht-Beste E, Ammitzbol Lauridsen C, Bachmann Nielsen M, et al. The performance of deep learning algorithms on automatic pulmonary nodule detection and classification tested on different datasets that are not derived from LIDC-IDRI: a systematic review. *Diagnostics* 2019;9:207.
- [66] Ardila D, Kiraly AP, Bharadwaj S, Choi B, Reicher JJ, Peng L, et al. End-to-end lung cancer screening with three-dimensional deep learning on low-dose chest computed tomography. *Nat Med* 2019;25:954.
- [67] Huang P, Lin CT, Li YL, Tammemagi MC, Brock MV, Atkar-Khattara S, et al. Prediction of lung cancer risk at follow-up screening with low-dose CT: a training and validation study of a deep learning method. *Lancet Digit Health* 2019;1:e353–62.
- [68] Jacobs C, Setio AAA, Scholten ET, Gerke PK, Bhattacharya H, Hoesein FAM, et al. Deep learning for lung cancer detection on screening CT: results of a large-scale public competition and an observer study with 11 radiologists. *Radiol Artif Intell* 2021;3:e210027.
- [69] Couraud S, Ferretti G, Milleron B, Cortot A, Girard N, Gounant V, et al. Intergroupe Francophone de Cancérologie Thoracique, Société de Pneumologie de langue française, and Société d'Imagerie Thoracique statement paper on lung cancer screening. *Diagn Interv Imaging* 2021;102:199–211.
- [70] Howlader N, Forjaz G, Mooradian MJ, Meza R, Kong CY, Cronin KA, et al. The effect of advances in lung-cancer treatment on population mortality. *N Engl J Med* 2020;383:640–9.
- [71] de Margerie-Mellon C, Gill RR, Salazar P, Oikonomou A, Nguyen ET, Heidinger BH, et al. Assessing invasiveness of subsolid lung adenocarcinomas with combined attenuation and geometric feature models. *Sci Rep* 2020;10:14585.
- [72] Yuan M, Zhang YD, Pu XH, Zhong Y, Li H, Wu JF, et al. Comparison of a radiomic biomarker with volumetric analysis for decoding tumour phenotypes of lung adenocarcinoma with different disease-specific survival. *Eur Radiol* 2017;27:4857–65.
- [73] Fan L, Fang MJ, Li ZB, Tu WT, Wang SP, Chen WF, et al. Radiomics signature: a biomarker for the preoperative discrimination of lung invasive adenocarcinoma manifesting as a ground-glass nodule. *Eur Radiol* 2019;29:889–97.
- [74] Huang HZ, Zheng DZ, Chen H, Wang Y, Chen C, Xu LC, et al. Fusion of CT images and clinical variables based on deep learning for predicting invasiveness risk of stage I lung adenocarcinoma. *Med Phys* 2022;49:6384–94.
- [75] Gong J, Liu JY, Li HM, Zhu H, Wang TT, Hu TD, et al. Deep learning-based stage-wise risk stratification for early lung adenocarcinoma in CT images: a multicenter study. *Cancers* 2021;13:3300.
- [76] Zhong YF, She YL, Deng JJ, Chen SY, Wang TT, Yang ML, et al. Deep learning for prediction of N2 metastasis and survival for clinical stage I non-small cell lung cancer. *Radiology* 2022;302:200–11.
- [77] Rios Velazquez E, Parmar C, Liu Y, Coroller TP, Cruz G, Stringfield O, et al. Somatic mutations drive distinct imaging phenotypes in lung cancer. *Cancer Res* 2017;77:3922–30.
- [78] Wang S, Shi JY, Ye ZX, Dong D, Yu DD, Zhou M, et al. Predicting EGFR mutation status in lung adenocarcinoma on computed tomography image using deep learning. *Eur Respir J* 2019;53:1800986.
- [79] Wang S, Yu H, Gan YC, Wu ZJ, Li EC, Li XH, et al. Mining whole-lung information by artificial intelligence for predicting EGFR genotype and targeted therapy response in lung cancer: a multicohort study. *Lancet Digit Health* 2022;4:e309–19.
- [80] He BX, Dong D, She YL, Zhou CC, Fang MJ, Zhu YB, et al. Predicting response to immunotherapy in advanced non-small-cell lung cancer using tumor mutational burden radiomic biomarker. *J ImmunoTher Cancer* 2020;8:e000550.
- [81] Hosny A, Parmar C, Coroller TP, Grossmann P, Zeleznik R, Kumar A, et al. Deep learning for lung cancer prognostication: a retrospective multi-cohort radiomics study. *PLoS Med* 2018;15:e1002711.
- [82] Kim H, Goo JM, Lee KH, Kim YT, Park CM. Preoperative CT-based deep learning Model for predicting disease-free survival in patients with lung adenocarcinomas. *Radiology* 2020;296:216–24.
- [83] Chassagnon G, Dohan A. Artificial intelligence: from challenges to clinical implementation. *Diagn Interv Imaging* 2020;101:763–4.
- [84] Gong B, Soyer P, McInnes MDF, Patlas MN. Elements of a good radiology artificial intelligence paper. *Can Assoc Radiol J* 2022. doi: 10.1177/08465371221101195.
- [85] Mongan J, Moy L, Kahn CE. Checklist for artificial intelligence in medical imaging (CLAIM): a guide for authors and reviewers. *Radiol Artif Intell* 2020;2:e200029.
- [86] Hosny A, Parmar C, Quackenbush J, Schwartz LH, Aerts H. Artificial intelligence in radiology. *Nat Rev Cancer* 2018;18:500–10.
- [87] Alexander R, Waite S, Bruno MA, Krupinski EA, Berlin L, Macknik S, et al. Mandating limits on workload, duty, and speed in radiology. *Radiology* 2022;212:631.
- [88] Hoang-Thi TN, Vakalopoulou M, Christodoulidis S, Paragios N, Revel MP, Chassagnon G. Deep learning for lung disease segmentation on CT: which reconstruction kernel should be used? *Diagn Interv Imaging* 2021;102:691–5.
- [89] Soyer P, Fishman EK, Rowe SP, Patlas MN, Chassagnon G. Does artificial intelligence surpass the radiologist? *Diagn Interv Imaging* 2022;103:445–7.
- [90] Waymel Q, Badr S, Demondion X, Cotten A, Jacques T. Impact of the rise of artificial intelligence in radiology: what do radiologists think? *Diagn Interv Imaging* 2019;100:327–36.
- [91] Chassagnon G, De Margerie-Mellon C, Vakalopoulou M, Marini R, Hoang-Thi TN, Revel MP, Soyer P. Artificial intelligence in lung cancer: current applications and perspectives. *Jpn J Radiol* 2022. doi: 10.1007/s11604-022-01359-x.